APOLLO 13

Apollo 13 was an American space mission in 1970, part of the Apollo program. It was intended to be the third mission to land on the Moon.

An oxygen tank in the Service Module exploded while the spacecraft was on its way to the Moon, requiring the mission to be aborted: the Moon landing was cancelled and only a single pass around the Moon was made. Considerable ingenuity under extreme pressure was required from both the crew and the ground controllers to figure out how to jury-rig the craft for the crew's safe return, with much of the world watching the drama on television.

Reentry into the Earth's atmosphere required the unusual step of undocking the lunar module, which had been retained for the flight back to Earth, in addition to the separation of the damaged service module. The lunar module had remained attached to the spacecraft to provide emergency propulsion and life support. The crew returned unharmed to the Earth.

In the wake of the near-disaster, NASA appointed a review board under the leadership of Edgar M. Cortright, director of Langley Research Center, to investigate the Apollo 13 accident. After some three months of study, the cause of the explosion was traced to two inadequate thermostatic switches in an oxygen-tank heater assembly. Dysfunction of the switches under load caused overheating that led to an insulation fire, and the subsequent blast tore a side panel from the service module and disabled the fuel cells. Other defects in manufacture and in testing procedures were also found. Further Apollo flights were delayed until 1971 so that modifications could be made to prevent similar incidents.

Jim Lovell, who was one of the crew members, wrote a book about the mission, *Lost Moon,* which was later turned into a successful movie, "Apollo 13", starring Tom Hanks.

EXERCISE 1: Find words or phrases in the passage which mean the same as:

	COLUMN A	COLUMN B
a)	stop before something is completed	
b)	large or relatively large in amount or degree	
c)	skill in inventing things or in working out plans	
d)	workout *(phrase)*	
e)	fix something that has broken away temporarily using whatever materials are available, but not necessarily the right materials	
f)	(of spaceships) the separation of a part of a spaceship in space	
g)	continue to keep something, especially when useful	
h)	power that moves something, especially a vehicle, in a forward direction	
i)	following after something, often as a result of the first thing *(phrase)*	
j)	a group of experts who meet for a common purpose	
k)	go back over again to discover the reason for something	
l)	not enough; insufficient; not good enough	
m)	a small control for an electrical device which you use to operate it	
n)	something which has been fitted together as part of a machine	
o)	incorrect operation or damaged in operation	
p)	a thick layer of material that is used to stop heat escaping from something	
q)	happening or existing at a later time than something else	
r)	a large violent explosion	
s)	fault orweakness	

EXERCISE 2: Choose the correct answer according to the passage.

1. According to the passage, the plan to return the Apollo 13's crew safely to the Earth.....

A) failed to save all the crew
B) was defective and amateurish
C) required inventive skills
D) was led by Edgar M. Cortright
E) was put off until 1971

2. As was stated in the passage, Jim Lovell ..... .

A) managed to return to the Earth, but with injuries
B) starred in a movie about the Apollo 13 mission
C) led the investigation into the near disaster
D) is an author with no first-hand knowledge of space travel
E) was among the crew of the Apollo 13

3. According to the details in the passage, the Apollo 13 mission could be described as.....

A) a perfect example for future space missions to follow
B) a successful attempt to land for the third time on the Moon
C) a planned space mission that never left the ground
D) a dramatic, eventful, but unsuccessful mission to land on the Moon
E) the first space mission broadcast live on television

EXERCISE 3: Complete the sentences by selecting words from Column B in EXERCISE 1.

1. A team of commandos was to fly secretly to the area to free the hostages, but the action was .....when the negotiations began producing some results.
2. A student who had been travelling on the coach showed remarkable..... when he suggested using his mother's nylon tights as a temporary belt for the engine. This temporary belt allowed the coach to reach the nearest town where proper repairs could be carried out.
3. The government investigation board.....the cause of the train crash to some loose bolts on the railway line.
4. The mechanic said that there was a fault on a switch in the air conditioning .....on my car and that, unfortunately, Renault only sold complete air conditioning systems.
5. Air travel in the United States declined dramatically.....the terrorist attacks of 11 September 2001.